ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

ENGINEERING MATHEMATICS I

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 1st year, semester 1 final examination: algebra and functions, complex numbers, limits (including L'Hospital's rule) and differentiation, integration (including improper integrals and Taylor series), matrices, vectors, lines and planes.

Written by Academic Central for practice, following first-year Engineering Mathematics syllabuses at South African universities (BEng Mechatronics, Electrical and Mechanical Engineering). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Show ALL your working. Marks are given for correct steps, not only for final answers.
3. Non-programmable scientific calculators may be used.
4. Give answers correct to TWO decimal places unless the question says otherwise. Angles are in degrees unless radians are asked for.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: ALGEBRA AND FUNCTIONS

1.1 Solve for x: $\log_2 x + \log_2 (x - 2) = 3$ (4)

1.2 Express $(3x + 5) / ((x + 1)(x + 3))$ in partial fractions. (4)

1.3 Given $f(x) = \sqrt{(x - 2)}$:

1.3.1 Write down the domain and the range of f. (2)

1.3.2 Find $f^{-1}(x)$ and state its domain. (2)

1.4 Express $3 \sin\theta + 4 \cos\theta$ in the form $R \sin(\theta + \alpha)$, with $R > 0$ and $0° < \alpha < 90°$. Hence write down the maximum value of $3 \sin\theta + 4 \cos\theta$. (3)

[15]

QUESTION 2: COMPLEX NUMBERS

Let $z_1 = 3 + 4j$ and $z_2 = 1 - 2j$, where $j^2 = -1$.

2.1 Calculate, giving each answer in the form $a + bj$:

2.1.1 $z_1 + z_2$ (1)

2.1.2 $z_1 \cdot z_2$ (2)

2.1.3 z_1 / z_2 (3)

2.2 Write z_1 in polar form $r\angle\theta$. (2)

2.3 Use De Moivre's theorem to calculate $(1 + j)^8$. (3)

2.4 Solve $z^2 + 2z + 5 = 0$. (2)

2.5 Write the complex number $2\angle 90°$ in exponential (Euler) form and in the form $a + bj$. (2)

[15]

QUESTION 3: LIMITS AND DIFFERENTIATION

3.1 Evaluate the following limits:

3.1.1 $\lim (x \to 2)$ of $(x^2 - 4) / (x - 2)$ (2)

3.1.2 $\lim (x \to \infty)$ of $(3x^2 + 1) / (2x^2 - x)$ (2)

3.1.3 Use L'Hospital's rule to evaluate $\lim (x \to 0)$ of $(e^x - 1 - x) / x^2$ (2)

3.2 Use the definition of the derivative (first principles) to find f′(x) if f(x) = x². (4)

3.3 Find dy/dx for each of the following:

3.3.1 $y = x^3 e^{2x}$ (3)

3.3.2 $y = \ln(x^2 + 1)$ (2)

3.3.3 $y = (\sin x) / x$ (3)

3.4 Use implicit differentiation to find the gradient of the circle $x^2 + y^2 = 25$ at the point (3; 4). (3)

3.5 An open box is made from a square sheet of metal 12 cm by 12 cm by cutting equal squares of side x cm from each corner and folding up the sides. Find the value of x that gives the largest volume, and calculate that volume. Show that it is a maximum. (4)

[25]

QUESTION 4: INTEGRATION

4.1 Find $\int (3x^2 - 4x + 5)\,dx$ (2)

4.2 Use a suitable substitution to find $\int 2x\,(x^2 + 1)^4\,dx$ (3)

4.3 Use integration by parts to find $\int x\,e^x\,dx$ (4)

4.4 Evaluate the improper integral $\int$ from 1 to ∞ of $1/x^2\,dx$ (2)

4.5 Find the area of the region enclosed between the curves $y = x^2$ and $y = 2x$. (5)

4.6 Use your answer to QUESTION 1.2 to find $\int (3x + 5) / ((x + 1)(x + 3))\,dx$ (3)

4.7 The region under $y = \sqrt{x}$ from x = 0 to x = 4 is rotated about the x-axis. Calculate the volume of the solid formed. Leave your answer in terms of π. (3)

4.8 Write down the Maclaurin (Taylor) series of e^x up to the term in x^3, and use it to estimate e^0.1 correct to FOUR decimal places. (3)

[25]

QUESTION 5: MATRICES AND VECTORS

Let A = [2 1 ; 1 3] and B = [1 −1 ; 0 2] (the rows of each matrix are separated by a semicolon).

5.1 Calculate AB. (2)

5.2 Calculate det A and find A^{-1}. (3)

5.3 Use A^{-1} to solve the system of equations: (3)

2x + y = 5

x + 3y = 10

5.4 Calculate the determinant of the matrix [1 2 0 ; 3 −1 4 ; 2 0 1]. (3)

5.5 Given the vectors a = 2i − j + 2k and b = i + 2j − 2k:

5.5.1 Calculate a · b. (2)

5.5.2 Calculate the angle between a and b. (2)

5.5.3 Calculate a × b. (3)

5.5.4 Find the equation of the plane that passes through the point (1; 0; 2) and has a × b as its normal vector. (2)

[20]

TOTAL: 100

END OF QUESTION PAPER